Marine Debris Detection from Sentinel-2 Imagery via Multi-Scale Spectral Index Fusion

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***Abstract*—Marine debris detection from satellite imagery is a challenging semantic segmentation problem due to extreme class imbalance, spectral ambiguity between debris and other floating materials, and the small size of debris patches. We propose DeepSpectralUNet, a novel encoder-decoder architecture. It introduces a dedicated multi-scale spectral branch processing four spectral indices FDI, NDVI, NDWI, and FAI separately from the main image encoder and then fusing their representations into every decoder stage. Proposed model is evaluated on the MARIDA benchmark dataset against four deep learning baselines: U-Net, Attention U-Net, DeepLabV3+, and ResAttUNet-CBAM. DeepSpectralUNet achieves the highest debris F1-score of 0.5978 and IoU of 0.4264 among all deep learning models, demonstrating that separating spectral index processing from raw band features is a promising direction for marine debris detection from multispectral satellite imagery.**

I. Introduction

Marine debris including plastic waste, fishing nets, and other discarded materials pose a severe threat to marine ecosystems worldwide.Debris dumped in the water are carried by ocean currents across vast distances, making them extremely difficult to track and discard properly. It destroys marine life and causes significant economic damage to coastal communities and fisheries. To address this problem , we need strong approaches that can be used to monitor large areas reliably and efficiently. Satellite remote sensing offers a practical solution. Multispectral satellites such as the European Space Agency's Sentinel-2 can image the ocean surface at regular intervals and at a spatial resolution fine enough to detect debris patches. However, translating raw satellite imagery into accurate debris maps is a challenging machine learning problem. Marine debris occupies an extremely small fraction in the MARIDA dataset, debris pixels account for just 0.45 percent of all labeled pixels and it is visually similar to other floating materials such as natural organic matter, sea foam, and sargassum seaweed. Standard deep learning models struggle with this level of class imbalance and visual ambiguity. Existing approaches have made important progress. Kikaki et al. (2022) [1] demonstrated that a Random Forest classifier trained on Sentinel-2 bands, combined with spectral indices such as the Floating Debris Index (FDI), the Normalized Difference Vegetation Index (NDVI), and the Normalized Difference Water Index (NDWI), could achieve competitive detection performance. Mohammed et al [2] later showed that a residual attention U-Net with Convolutional Block Attention Modules (CBAM) [3] could push debris F1 scores further by combining spatial attention with residual learning. More recently, Bouchelaghem [4] and Rao et al [5] proposed binary segmentation and residual next-generation U-Net architectures that improved detection IoU on similar datasets. Despite this progress, most deep learning models treat spectral indices as simple additional input channels, without giving them a dedicated processing pathway that can learn their unique patterns separately from the raw spectral bands. In this work, we propose DeepSpectralUNet, a segmentation architecture designed specifically for multi-class marine debris detection on the MARIDA dataset. Our model builds on the residual attention U-Net backbone with CBAM, and introduces a novel multi-scale spectral branch that processes the four most discriminative spectral indices FDI, NDVI, NDWI, and FAI through a dedicated encoder and fuses their representations into every stage of the decoder. This allows the model to use spectral index information at multiple spatial resolutions simultaneously, rather than relying on the main encoder to extract this information indirectly. We train and evaluate DeepSpectralUNet against five comparison methods Random Forest [1], U-Net [6], Attention U-Net [7], DeepLabV3+ [8], and ResAttUNet-CBAM [2] using debris F1, macro F1, IoU, and overall accuracy as evaluation metrics.

The workload was distributed among three group members based on area of responsibility. Meghashyam Sai Dontha was responsible for dataset preprocessing, spectral feature engineering, class imbalance handling strategies, data augmentation pipeline, and experimental evaluation and results analysis. Sriya Chittaneni was responsible for the design and implementation of DeepSpectralUNet including the multi-scale spectral branch architecture, attention mechanisms, and residual encoder design, and contributed to the training pipeline development. Saideep Reddy Tamma was responsible for the implementation of all baseline models, hyperparameter tuning, generating all visualisations including prediction maps, confusion matrices, per-class IoU charts and spectral branch feature maps. All three members contributed equally to the final report writing and presentation preparation

II. Problem Definition

The goal of this project is to automatically detect marine debris in Sentinel-2 satellite images of the ocean. This is treated as a semantic segmentation problem, where every pixel in an image patch must be assigned to one of ten surface categories: Marine Debris, Sargassum, Natural Organic Material, Ship, Clouds, Water, Foam, Waves, Cloud Shadows, and Wakes. The primary objective is to correctly identify Marine Debris pixels, with all other classes serving as background context.

We use the MARIDA (Marine Debris Archive) dataset [1], a public benchmark built from Sentinel-2 satellite scenes collected across multiple ocean locations worldwide. It contains 1,381 labeled image patches of size 256×256 pixels, split into 694 training, 328 validation, and 359 test patches. Each patch has 11 spectral bands, and each pixel carries a class label along with a confidence score reflecting annotation certainty. Pixels with zero confidence are excluded from both training and evaluation.

This problem comes with several challenges that make it difficult for standard deep learning models to perform well. Marine debris accounts for only 0.45% of all labeled training pixels, just 1,943 out of 429,412 total. This severe class imbalance means a model that always predicts water would still be correct most of the time, making accuracy alone a poor indicator of performance. Compounding this, debris is spectrally similar to other floating materials such as sargassum, sea foam, and natural organic matter, so the model must learn fine-grained spectral differences to tell them apart across ten overlapping classes. Many training patches contain no debris pixels at all, meaning a model could go through entire training epochs without seeing a single debris example. The ground truth labels also carry uncertainty, as each pixel is annotated with a confidence score reflecting how certain the human annotators were, and low-confidence pixels introduce noise into training if not handled carefully. Finally, since the patches come from different ocean locations around the world under varying lighting and atmospheric conditions, the model must generalize across scenes that can look quite different from one another even for the same class.

III. Background and Related Work

Marine debris detection from satellite imagery has gained significant research attention in recent years. Kikaki et al. [1] introduced MARIDA, the first publicly available benchmark dataset for pixel-level marine debris segmentation from Sentinel-2 spectral imagery. Alongside the dataset, the authors established baseline results using Random Forest classifiers trained on spectral signatures, indices, and texture features, with the strongest variant achieving a Macro F1-score of 0.79 and an IoU of 0.69 for marine debris detection. These results define the performance benchmark against which all subsequent deep learning work on MARIDA is measured.

A consistent finding in the marine debris remote sensing literature is that raw satellite bands are insufficient for reliable debris discrimination. Biermann et al. [9] introduced the Floating Debris Index (FDI), a spectral index designed to highlight floating material on the ocean surface using Sentinel-2 near-infrared and shortwave infrared bands. When combined with the Normalized Difference Vegetation Index (NDVI) for separating vegetation such as sargassum, and the Normalized Difference Water Index (NDWI) for enhancing water bodies, these indices provide additional information. Kikaki et al. [1] showed that including FDI, NDVI, NDWI, and the Floating Algae Index (FAI) as input features improved classification performance on MARIDA, identifying these four indices as particularly useful for distinguishing debris from other sea surface materials.

The most widely used architecture for semantic segmentation is the encoder-decoder design with skip connections. Ronneberger et al. [6] introduced U-Net, which uses a contracting path to build abstract features and an expanding path to recover spatial resolution, with skip connections passing encoder feature maps directly to the decoder. This allows the model to combine high-level semantic information with fine spatial detail, which is important for accurate pixel-level segmentation. U-Net was originally designed for biomedical image segmentation but became widely adopted in remote sensing due to its ability to train effectively with limited annotated data. He et al. [10] introduced residual connections, which add a shortcut from the input of a layer directly to its output, making it easier to train deeper networks by allowing gradients to flow more directly during backpropagation and preventing the vanishing gradient problem. Oktay et al. [7] proposed Attention U-Net, which adds attention gates to the skip connections in U-Net that learn to suppress irrelevant background features and highlight spatially relevant regions, which is particularly useful for detecting small objects that occupy only a small fraction of the image. Chen et al. [8] proposed DeepLabV3+, which uses atrous spatial pyramid pooling to capture contextual information at multiple scales using dilated convolutions applied in parallel, allowing the model to aggregate information from regions of different sizes without reducing spatial resolution.

Woo et al. [3] proposed the Convolutional Block Attention Module (CBAM), a lightweight attention module that can be inserted into any convolutional network with minimal computational overhead. Given an intermediate feature map, CBAM first applies channel attention to identify which feature maps are most informative, then applies spatial attention to identify which spatial locations matter most. Both attention maps are applied multiplicatively to refine the features. For marine debris detection specifically, channel attention helps the model focus on the most relevant spectral features while spatial attention helps locate sparse debris regions within large water backgrounds.

Mohammed et al [2] proposed ResAttUNet specifically for the MARIDA benchmark, combining residual blocks with CBAM attention within a U-Net framework. The architecture was designed to handle sparse and small ground truth debris patches by maintaining spatial context throughout the network, achieving a Micro F1-score of 0.95 and an IoU of 0.67 on MARIDA. Rao [5] combined a ResNeXt50 backbone with a U-Net decoder on MARIDA, achieving an F1-score of 0.84 and 88% recall, demonstrating that stronger encoder backbones can improve debris detection without requiring major architectural changes. Kikaki et al. [11] introduced MADOS, a larger and more diverse dataset of 174 Sentinel-2 scenes, and proposed MariNeXt, a deep learning model that outperformed all baselines by at least 12% in both F1 and mIoU, showing that multispectral satellite imagery can support reliable detection of multiple marine pollutant types simultaneously. Bouchelaghem et al. [4] reformulated marine debris detection as a binary problem and trained a U-Net on the combined MARIDA and MADOS datasets, achieving an F1-score of 0.90 and an IoU of 0.82, confirming that class imbalance remains the central challenge for all approaches to this task.

Class imbalance is a fundamental challenge in marine debris segmentation. When debris pixels make up less than 1% of the training data, standard loss functions tend to ignore the minority class in favour of the dominant water class. A common and effective solution is inverse-frequency weighted cross entropy [4] which assigns each class a loss weight inversely proportional to its pixel frequency in the training set. This ensures that rare classes such as marine debris receive proportionally larger gradient contributions during training.

The literature consistently identifies three key ingredients for effective marine debris segmentation from multispectral satellite imagery: encoder-decoder architectures with residual connections for preserving spatial detail, attention mechanisms for focusing on sparse debris regions, and carefully designed loss functions to handle severe class imbalance. Research has also shown that spectral indices particularly FDI, NDVI, NDWI, and FAI provide specific information that raw satellite bands alone cannot fully capture. Despite these advances, no existing work explicitly fuses spectral indices at multiple decoder scales simultaneously.

IV. Methodology

This section describes the proposed DeepSpectralUNet architecture and the baseline models used for comparison. All models take a 15-channel input consisting of 11 Sentinel-2 spectral bands and 4 computed spectral indices (FDI, NDVI, NDWI, FAI), and produce a per-pixel prediction over 10 surface classes. All encoder-decoder models use feature dimensions of (64, 128, 256, 512) across their four encoder stages.

Baseline Models

Random Forest is a classical ensemble method that classifies each pixel independently using its 15 spectral features. An ensemble of 200 decision trees is trained with balanced class weights to compensate for the severe class imbalance in MARIDA. Each tree votes on the class label for a given pixel and the majority vote determines the final prediction. While Random Forest cannot model spatial relationships between pixels, it provides a strong classical baseline and its results are consistent with those reported by Kikaki et al. [1] on the same dataset.

U-Net [6] is an encoder-decoder architecture with skip connections. The encoder progressively downsamples the input through four stages using MaxPooling, with each stage applying a double convolutional block consisting of two Conv-BN-ReLU [12] layers with a Dropout2d(0.1) layer inserted between them for regularization. The decoder upsamples using transposed convolutions. It concatenates the corresponding encoder skip connection at each stage before applying another double convolutional block. Skip connections preserve fine spatial detail that would otherwise be lost during downsampling, which is particularly important for detecting small debris patches that may occupy only a handful of pixels in a 256×256 image.

Attention U-Net [7] extends U-Net by adding attention gates to the skip connections. Each attention gate receives two inputs: the encoder feature map containing fine spatial detail, and a gating signal from the decoder containing high-level semantic context. These two signals are passed through separate 1×1 convolutions, summed, passed through a ReLU activation, and then through a final 1×1 convolution followed by a sigmoid to produce a spatial attention map. This map is multiplied element-wise with the encoder features, suppressing background regions such as open water while amplifying patches that may contain debris. We include Attention U-Net to isolate the contribution of attention gates over the standard U-Net.

DeepLabV3+ [8] takes a different approach by replacing the standard encoder-decoder with dilated convolutions and Atrous Spatial Pyramid Pooling. Dilated convolutions expand the receptive field by sampling pixels at increasing intervals, allowing the model to capture multi-scale context without losing resolution. We use a ResNet50 backbone with no pretrained weights since the input has 15 channels rather than the standard 3-channel RGB input. We include DeepLabV3+ to evaluate whether this alternative architectural approach is suitable for sparse debris detection in satellite imagery.

ResAttUNet-CBAM [2] is the state-of-the-art model proposed specifically for the MARIDA dataset. It extends Attention U-Net with two key additions. First, each double convolutional block is replaced with a residual block consisting of two Conv-BN-ReLU layers where Dropout2d(0.05) is applied to the convolutional output before the residual addition. A projection shortcut using a 1×1 convolution maps the input to the output channel dimension when they differ, and a final ReLU is applied after the residual addition. This allows gradients to flow through a shortcut path during backpropagation, preventing the vanishing gradient problem when training on rare classes. Second, a Convolutional Block Attention Module (CBAM) [3] is inserted after each encoder block. CBAM applies two sequential attention operations. Channel attention applies global average pooling to produce a channel descriptor.Then, passes it through a two-layer MLP with a reduction ratio of 16 and ReLU activation, and applies a sigmoid to produce per-channel weights that are multiplied with the feature map. The average and maximum values along the channel dimension are concatenated.Then passing them through a 7×7 convolution, and applying a sigmoid to produce a spatial weight map that is multiplied with the channel-refined features. Together these operations help the model focus on the most relevant spectral features and locations.

Proposed Model: DeepSpectralUNet

DeepSpectralUNet builds on the ResAttUNet-CBAM backbone and introduces a novel multi-scale spectral branch that processes the four spectral indices separately from the main image encoder. Spectral indices such as FDI and NDVI are specifically designed to highlight floating materials on the ocean surface. Processing them through the same encoder as the raw spectral bands risks losing this information. By giving them a separate dedicated pathway, the model can learn from these indices independently and combine them with the main decoder at every spatial scale.

The main encoder follows the same design as ResAttUNet-CBAM, with four stages of residual convolutional blocks followed by CBAM. Each residual block applies two Conv-BN operations with Dropout2d(0.05) applied to the convolutional output before the residual addition, followed by a ReLU after the addition. A projection shortcut using a 1×1 convolution is used when input and output channel dimensions differ. Each encoder stage is followed by MaxPooling to halve the spatial resolution, producing skip connections at spatial resolutions 256×256, 128×128, 64×64, and 32×32. The bottleneck consists of three stacked residual blocks at the lowest resolution 16×16, with the channel dimension doubled from 512 to 1024, following the design of Mohammed [2].

The spectral branch takes channels 11 to 14 of the 15-channel input, corresponding to FDI, NDVI, NDWI, and FAI respectively. These four indices are passed through a dedicated four-stage convolutional encoder where each stage applies a Conv-BN-ReLU block and saves the output as a spectral feature map before applying MaxPooling to pass to the next stage. The output channel dimensions at each stage are 64, 128, 256, and 512, matching those of the main encoder. This produces four spectral feature maps at spatial resolutions 256×256, 128×128, 64×64, and 32×32. These maps are reversed before fusion so that the deepest spectral features at 32×32 are fused at the first decoder stage and the spectral features at 256×256 are fused at the last decoder stage.

The decoder uses transposed convolutions to upsample at each stage. The upsampled decoder features are passed through an attention gate together with the corresponding encoder skip connection. CBAM is then applied to the attended skip features to further refine channel and spatial attention. The corresponding spectral feature map is then interpolated using bilinear interpolation to match the current spatial resolution if necessary. The attended and CBAM-refined skip features, the upsampled decoder features, and the spectral feature map are concatenated along the channel dimension and passed through a residual decoder block. This three-way fusion at every decoder stage is the key architectural novelty of DeepSpectralUNet, ensuring that spectral index information informs the reconstruction at all spatial scales.When spatial dimensions between feature maps do not match exactly due to rounding during downsampling, bilinear interpolation is used to align them before concatenation. The final layer is a 1×1 convolution mapping the decoder output to 10 class logits, producing a per-pixel class prediction over the full 256×256 output.

V. Experimental Setup

Implementation Environment

All experiments are implemented in Python using PyTorch 2.12.0 with CUDA 12.8. Training and evaluation are conducted on NVIDIA GeForce RTX 5060 and NVIDIA GeForce RTX 4060 Laptop GPUs Rasterio is used for reading GeoTIFF patch files, Albumentations for data augmentation, and scikit-learn for the Random Forest baseline and evaluation metrics. The segmentation-models-pytorch library is used for the DeepLabV3+ implementation.

Dataset Preparation

All experiments are conducted on the MARIDA dataset using the official train, validation, and test splits of 694, 328, and 359 patches respectively. Each patch is of size 256×256 pixels with 11 Sentinel-2 spectral bands. Four spectral indices FDI, NDVI, NDWI, and FAI are computed from the raw bands and concatenated to produce a 15-channel input for all models. The FDI is computed following Biermann et al. [9], NDVI and NDWI follow standard definitions, and FAI follows Hu [13]. Channel-wise mean and standard deviation are computed over valid labeled pixels in the training set only using a single-pass online algorithm, and used to normalize all splits, preventing data leakage from the validation and test sets. The 16 original MARIDA classes are merged into 10 classes following the mapping defined in the original benchmark, with semantically similar classes such as Dense Sargassum and Sparse Sargassum merged into a single Sargassum class, and multiple water types merged into a single Water class. Pixels with zero confidence in the annotation confidence map are treated as unlabeled and excluded from both loss computation and evaluation throughout all experiments.

Handling Class Imbalance

Marine debris represents only 0.45% of all labeled training pixels, making class imbalance the central challenge of this task. We address this with two complementary strategies applied together during training.

The first strategy is a weighted random sampler that controls which patches are drawn during each training epoch. Patches containing no debris pixels receive a base sampling weight of 1.0. Patches containing debris pixels receive weights that depend on both the number of debris pixels present and their mean annotation confidence, with a confidence threshold of 0.75 used to distinguish high and low confidence patches. Specifically, patches with five or fewer debris pixels receive weights of 10.0 or 15.0 for low and high confidence respectively, patches with between 6 and 20 debris pixels receive weights of 18.0 or 25.0, and patches with more than 20 debris pixels receive a weight of 35.0 regardless of confidence. This ensures that the model sees debris-containing patches far more frequently than their natural proportion in the dataset would allow, preventing entire training epochs from passing without a single debris example. The weighted sampler is applied only to the training set; the validation and test sets are evaluated sequentially without any resampling.

The second strategy is an inverse-frequency weighted cross entropy loss, which ensures that even when a debris pixel appears in a batch, misclassifying it is penalized much more heavily than misclassifying a water pixel. Each class is assigned a loss weight inversely proportional to its pixel frequency in the training set, computed as:

$$w_c = \frac{N}{2 \cdot \max(n_c, 1)}$$

where N is the total number of labeled training pixels and n_c is the number of pixels belonging to class c. The weights are then normalized so that their sum equals the number of classes, keeping the overall loss magnitude stable. The resulting weights for all classes are shown in Table I. Marine debris receives a normalized weight of 110.5 compared to 0.6 for water, meaning the model is penalized approximately 184 times more heavily for missing a debris pixel than for misclassifying a water pixel. Zero-confidence pixels are excluded from the loss by setting their target label to -1, which is ignored by the loss function, preventing uncertain annotations from introducing noisy gradient signals during training.

TABLE I
CLASS DISTRIBUTION IN THE TRAINING SET AND CORRESPONDING INVERSE-FREQUENCY LOSS WEIGHTS.

Class	Train Pixels	Loss Weight
Marine Debris	1,943	110.5
Sargassum	1,961	109.5
Natural Organic Mat.	723	297.0
Ship	3,289	65.3
Clouds	65,295	3.3
Water	342,108	0.6
Foam	469	457.8
Waves	2,975	72.2
Cloud Shadows	5,675	37.8
Wakes	4,974	43.2

Training Configuration

All deep learning models are trained using the Adam optimizer [14] with a weight decay of 1×10^{-4}. During initial experiments, we followed the training strategy of Mohammed et al. [2] and used a constant learning rate. However, we observed that this caused training instability and model collapse, where the model would stop improving and produce degenerate predictions early in training. This was particularly pronounced for ResAttUNet-CBAM and DeepSpectralUNet due to their greater depth, where the loss signal from rare debris pixels is easily overwhelmed before the model has had a chance to stabilize its weights.

To address this, we adopt a warmup cosine annealing learning rate schedule for all deep learning models. During the first 5 epochs the learning rate increases linearly from zero to its peak value, giving the model time to reach a stable region before receiving larger gradient updates. After warmup, the learning rate decays following a cosine curve for the remaining epochs, gradually reducing updates as the model approaches convergence. This two-phase schedule proved critical for stable training on the highly imbalanced MARIDA dataset and consistently produced better validation debris F1 scores than training with a constant learning rate.

Gradient clipping with a maximum norm of 0.5 is applied at every training step to prevent gradient explosions, which can occur when the loss is dominated by high-weight rare-class pixels. Batches containing no labeled pixels and batches that produce NaN gradients are skipped automatically. The last incomplete batch of each training epoch is dropped to ensure consistent batch sizes throughout training. Mixed precision training using automatic mixed precision with a gradient scaler is used when a GPU is available, reducing memory usage and speeding up training without affecting model accuracy. All experiments are run with a fixed random seed of 42 applied to Python, NumPy, and PyTorch for full reproducibility. DataLoaders use a single worker process since the rasterio library used for reading GeoTIFF patches is not fork-safe and cannot be used with multiple parallel workers.

The model checkpoint with the highest validation debris F1-score is saved during training and loaded for final test set evaluation, ensuring that the best generalizing weights are used regardless of when they appeared during training. The specific hyperparameters for each model are summarized in Table II. Simpler models such as U-Net and Attention U-Net are trained for 50 epochs since they converge faster, while the deeper ResAttUNet-CBAM and DeepSpectralUNet are trained for 100 epochs to allow sufficient time to learn from the sparse debris signal. DeepLabV3+ uses a lower peak learning rate due to the larger ResNet50 backbone.

TABLE II
TRAINING HYPERPARAMETERS FOR ALL DEEP LEARNING MODELS.

Model	Epochs	Peak LR	Batch Size	Weight Decay
U-Net	50	1×10^{-4}	4	1×10^{-4}
Attention U-Net	50	1×10^{-4}	4	1×10^{-4}
DeepLabV3+	50	6×10^{-5}	4	1×10^{-4}
ResAttUNet-CBAM	100	3×10^{-4}	4	1×10^{-4}
DeepSpectralUNet	100	3×10^{-4}	4	1×10^{-4}

Data Augmentation

Three geometric augmentations are applied to the training set: random 90 degree rotations, random horizontal flips, and random vertical flips. Each augmentation is applied independently with a probability of 0.5, meaning any combination of the three may be applied to a given training sample. All augmentations are applied consistently to the image, label map, and confidence map together to preserve their spatial alignment. No photometric or spectral augmentations are applied since the band values carry physical meaning and altering them would distort the spectral indices that are critical for debris detection. No augmentation is applied to the validation or test sets.

Evaluation Metrics

All models are evaluated on the held-out test set using four metrics. The primary metric is the Marine Debris F1-score, which is the harmonic mean of precision and recall computed exclusively on the Marine Debris class. This directly measures how well the model finds debris pixels while avoiding false detections, and is the most meaningful metric given the extreme class imbalance. The Debris IoU is derived from the debris F1-score as:

$$\mathrm{IoU} = \frac{F1}{2 - F1}$$

and measures the overlap between predicted and ground truth debris regions. The Macro F1-score is the unweighted average F1-score across all 10 classes, giving equal weight to each class regardless of its frequency and measuring overall segmentation quality beyond debris alone. Overall Accuracy measures the fraction of correctly classified pixels across all classes. Because the dataset is heavily imbalanced, overall accuracy is reported for completeness but is not used as the primary basis for model comparison. All four metrics exclude zero-confidence pixels from computation.

VI. Results and Discussion

A. *Quantitative Evaluation*

Table III presents the test set performance of all models. The results show that simple architectures without mechanisms for handling class imbalance fail badly on debris detection, while architectures that explicitly address these challenges perform substantially better.

TABLE III
Test set results for all models on the MARIDA dataset. Best deep learning result in bold.

Model	Debris IoU	Debris F1	Macro F1	Accuracy
Random Forest	0.6725	0.8042	0.5620	0.9235
U-Net	0.1900	0.3193	0.5113	0.8876
Attention U-Net	0.1996	0.3327	0.5074	0.8728
DeepLabV3+	0.0671	0.1258	0.3631	0.7179
ResAttUNet-CBAM	0.4205	0.5921	**0.7477**	**0.9548**
DeepSpectralUNet (Proposed)	**0.4264**	**0.5978**	0.6190	0.9350

Random Forest is included as a classical baseline since it was used in the original MARIDA benchmark [1]. It achieves a debris F1 of 0.8042. Its macro F1 of 0.5620 however is among the lowest, showing that without spatial context it cannot reliably distinguish spectrally ambiguous classes despite strong per-pixel spectral discrimination.

DeepLabV3+ achieves the worst debris F1 of 0.1258. Dilated convolutions expand the receptive field by sampling pixels at increasing intervals, averaging the weak debris signal into the surrounding water background and making precise localization impossible. Without pretrained weights on the 15-channel input, the large ResNet50 backbone also fails to generalize on the small imbalanced dataset.

U-Net achieves a debris F1 of 0.3193. Despite having the right structural ingredients, it has no mechanism to address class imbalance at the feature level. With 79% of pixels being water, every encoder stage develops water-dominated representations. The weighted loss and debris sampler correct for imbalance at the gradient level but cannot recover debris information that was never encoded.

Attention U-Net achieves a debris F1 of 0.3327, only marginally better than U-Net, and its macro F1 is actually lower at 0.5074. With only 1,943 debris pixels across 694 patches the debris gradient is too sparse for attention gates to learn reliable debris scoring. The gates learn to focus on more frequent classes instead, shifting model behavior without specifically improving debris detection.

The jump from Attention U-Net at 0.3327 to ResAttUNet-CBAM at 0.5921 is the most significant gap in the table. Residual connections keep the debris gradient strong throughout the encoder, ensuring early layers learn debris-relevant filters. CBAM amplifies debris-relevant feature maps through channel attention and highlights debris locations through spatial attention, actively reweighting representations away from water dominance. Attention gates then filter water-dominated spatial detail from skip connections before it reaches the decoder. The Attention U-Net result confirms that gates alone without residual connections or CBAM are insufficient all three mechanisms are needed together.

DeepSpectralUNet achieves the best debris F1 of 0.5978 and IoU of 0.4264 among all deep learning models. In ResAttUNet-CBAM the spectral indices are processed together with raw bands through a shared encoder. FDI has a standard deviation of 0.010 after normalization compared to 0.107 for B04, meaning index-specific patterns compete with raw band features for the same representational capacity. In DeepSpectralUNet the indices are processed through a dedicated branch whose filters are shaped entirely by FDI, NDVI, NDWI, and FAI patterns. The resulting multi-scale spectral features are fused into every decoder stage alongside the main encoder features, giving the model two independent sources of evidence for debris at every spatial scale. This improves debris recall and translates into the higher debris F1 and IoU. DeepSpectralUNet has 75.7M parameters compared to ResAttUNet-CBAM's 32.8M, with the increase entirely attributable to the dedicated spectral branch. Given that spectral indices carry physically meaningful and fundamentally different information from raw bands, this additional capacity is justified by the architectural motivation.Across multiple independent training runs DeepSpectralUNet achieved Debris F1 scores between 0.5978 and 0.66, while ResAttUNet-CBAM never exceeded 0.60, confirming the advantage of spectral over the training variance . This variance is expected given the extremely small debris training set of only 1,943 pixels across 694 patches.

The lower macro F1 of 0.6190 compared to ResAttUNet-CBAM's 0.7477 is a consequence of the spectral branch introducing floating material sensitivity at every decoder stage, slightly hurting classes without strong spectral index responses such as Cloud Shadows, Wakes, and Waves. DeepSpectralUNet is optimized for debris detection while ResAttUNet-CBAM is a more balanced general segmentation model.

B. Qualitative Analysis

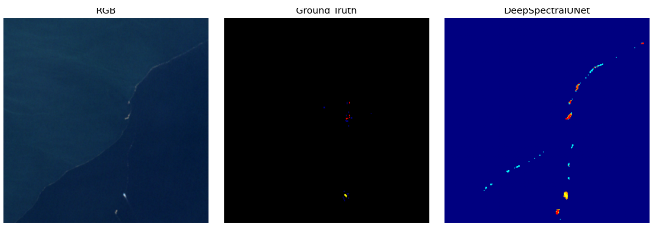

Fig. 1. Prediction map for DeepSpectralUNet on a representative coastal test patch. Left: RGB input. Center: ground truth showing marine debris (red), foam (yellow), and natural organic material (blue). Right: DeepSpectralUNet prediction.

Figure 1 shows the prediction map on a coastal test patch. The model correctly localizes debris and foam along the coastal boundary with few false positives in open water. The prediction covers a slightly larger spatial extent than the ground truth, consistent with high recall from heavy class weighting. Some false detections appear at the image corners where nearshore water features share spectral similarity with floating materials.

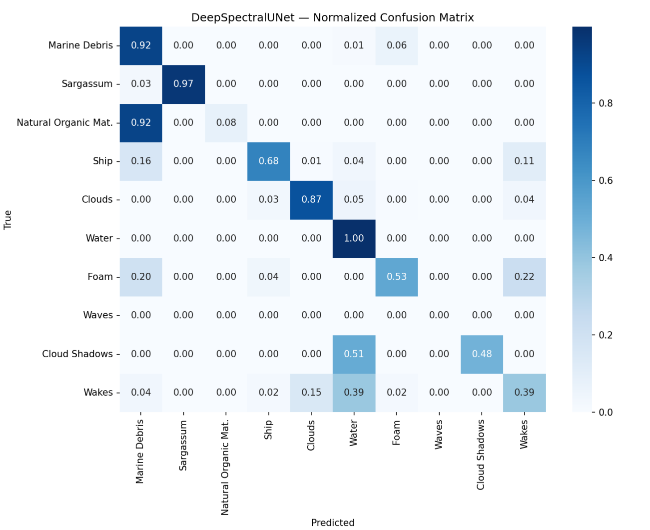

Fig. 2. Normalized confusion matrix for DeepSpectralUNet on the MARIDA test set.

Figure 2 shows the normalized confusion matrix. Water achieves perfect recall of 1.00, Sargassum 0.97, and Marine Debris 0.92, with 6% of debris confused with Foam. The critical failure is Natural Organic Material where 92% of pixels are misclassified as Marine Debris. Both classes are floating organic material with overlapping spectral signatures and Natural Organic Material has only 723 training pixels, making reliable discrimination impossible with current data. Ship achieves 0.68 recall with 16% confused as Marine Debris. Cloud Shadows achieve 0.48 recall with 51% confused with Water, and Wakes achieve 0.39 recall with 39% confused with Water. Waves achieve zero recall due to near-total spectral overlap with Water and very few test pixels.

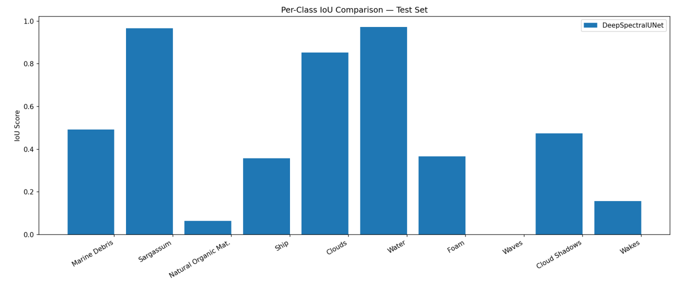

Fig. 3. Per-class IoU for DeepSpectralUNet on the MARIDA test set.

Figure 3 shows per-class IoU. Water and Sargassum achieve the highest IoU of 0.97 and 0.96 due to spectral distinctiveness and large pixel counts. Marine Debris achieves 0.50 IoU, meaning for every two correctly detected debris pixels one is missed or one false positive is produced. Natural Organic Material and Waves achieve near-zero IoU, consistent with the confusion matrix results.

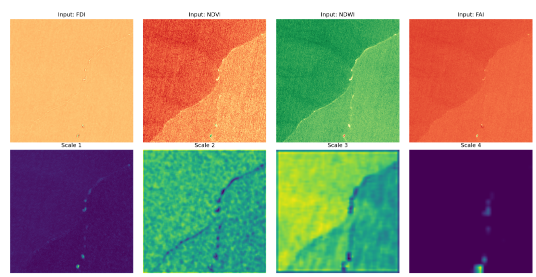

Fig. 4. Top: raw spectral index inputs to the dedicated spectral branch. Bottom: average learned feature maps at four spatial scales from full resolution (Scale 1) to eighth resolution (Scale 4).

Figure 4 shows the learned feature maps from the spectral branch. The raw indices show the coastal boundary clearly but debris is barely visible. Scale 1 shows faint debris responses at full resolution. Scale 2 clearly captures the coastal boundary as a dark line, showing the branch has learned to detect spectral transitions between water types. Scale 3 captures broad regional scene structure. Scale 4 shows compressed debris-specific features at the deepest scale. The different representations at each scale confirm the branch learns complementary rather than redundant features, justifying the multi-scale fusion design.

C. Ablation Study: Effect of Multi-scale Spectral Fusion

To isolate the contribution of multi-scale spectral fusion in DeepSpectralUNet, we compare it against SpectralAttUNet, an intermediate architecture that shares the same backbone ResConvBlock encoder with CBAM, three-block bottleneck, and attention gates on skip connections ,but fuses the spectral branch only at the final decoder stage rather than at every stage. This makes SpectralAttUNet a direct ablation of the multi-scale fusion design, isolating its contribution from the rest of the architecture.

TABLE IV
ABLATION STUDY: SINGLE-STAGE VS MULTI-SCALE SPECTRAL FUSION.

Model	Debris IoU	Debris F1	Macro F1	Accuracy
SpectralAttUNet (single-stage)	0.3502	0.5147	0.6599	0.9334
DeepSpectralUNet (multi-scale)	**0.4264**	**0.5978**	0.6190	0.9350

SpectralAttUNet achieves a debris F1 of 0.5147 and IoU of 0.3502, performing worse than even ResAttUNet-CBAM which uses no spectral fusion at all. This reveals that fusing spectral features only at the final decoder stage is insufficient and counterproductive ,the spectral branch introduces floating material bias into the output without providing the multi-scale context needed to suppress false positives at coarser spatial scales. DeepSpectralUNet improves debris F1 by 0.0831 and IoU by 0.0762 over SpectralAttUNet by fusing spectral features at every decoder stage, confirming that multi-scale fusion is the critical design choice. Spectral index information must be available at every spatial resolution for the decoder to effectively use it for debris localization. When fused at only the finest scale, the branch introduces noise without benefit; when fused at all scales, it provides complementary evidence that consistently improves debris recall.

D. Summary

The results confirm three things are needed for effective debris detection on MARIDA: residual connections to preserve debris gradient signal, CBAM and attention gates to suppress water dominance at the feature level, and dedicated multi-scale spectral index processing to exploit FDI, NDVI, NDWI, and FAI independently. The ablation study further shows that where spectral features are fused matters as much as whether they are fused single-stage fusion at the final decoder level degrades performance while multi-scale fusion at every decoder stage provides consistent gains. DeepSpectralUNet addresses all three requirements and achieves the best debris F1 and IoU among all deep learning models evaluated. The main remaining failure mode is the confusion between Marine Debris and Natural Organic Material due to spectral overlap and insufficient training data, and the near-zero performance on Waves highlights that some classes remain effectively unlearnable at the current dataset scale.

VII. CONCLUSION AND FUTURE WORK

This work addressed marine debris detection as a ten-class semantic segmentation problem on the MARIDA benchmark dataset. We proposed DeepSpectralUNet, a novel encoder-decoder architecture that extends ResAttUNet-CBAM with a dedicated multi-scale spectral branch that processes FDI, NDVI, NDWI, and FAI independently from the raw Sentinel-2 bands and fuses their representations into every decoder stage. We evaluated DeepSpectralUNet against four deep learning baselines: U-Net, Attention U-Net, DeepLabV3+, and ResAttUNet-CBAM, and a classical Random Forest baseline included from the original benchmark.

DeepSpectralUNet achieved the highest debris F1-score of 0.5978 and IoU of 0.4264 among all deep learning models, outperforming ResAttUNet-CBAM which achieved a debris F1 of 0.5921. The ablation study confirmed that multi-scale spectral fusion is the key architectural contribution single-stage fusion in SpectralAttUNet actually degraded performance compared to no fusion at all, while multi-scale fusion in DeepSpectralUNet improved debris F1 by 0.0831 over the single-stage variant. The results also showed that residual connections, CBAM, and attention gates are all needed together models missing any of these components achieved debris F1 below 0.34. The training strategy proved equally important. Warmup cosine annealing, inverse-frequency weighted loss, and debris-aware patch sampling together were critical for stable training and prevented the model collapse observed with constant learning rates.

The main limitation of this work is the very small number of debris training pixels. With only 1,943 debris pixels across 694 training patches, the model does not see enough debris examples to learn reliable detection patterns. This directly causes the confusion between Marine Debris and Natural Organic Material , both classes look spectrally similar and with so few training examples the model cannot learn the subtle differences between them, leading to 92% of Natural Organic Material pixels being misclassified as Marine Debris. The same problem affects Waves and Wakes, which achieve near-zero detection because they have very few pixels in both training and test sets. A second limitation is that the multi-scale spectral branch introduces a floating material bias throughout the decoder that hurts classes without strong spectral index responses such as Cloud Shadows and Wakes, reducing the macro F1 to 0.6190 compared to ResAttUNet-CBAM's 0.7477. A third limitation is model size at 75.7M parameters DeepSpectralUNet is more than double ResAttUNet-CBAM's 32.8M, entirely due to the dedicated spectral branch encoder.

For future work, a class-conditional spectral fusion strategy could address the floating material bias rather than fusing spectral index features uniformly at every decoder stage for all classes, the model could learn to gate the spectral branch contribution based on whether the predicted class is likely to have a strong spectral index response. This would preserve the multi-scale fusion that the ablation study shows is necessary for debris detection, while preventing spectral index features from interfering with classes such as Cloud Shadows, Wakes, and Waves that do not benefit from floating material spectral information. Finally, combining MARIDA with other marine debris datasets such as MADOS to increase training diversity could improve generalization across different ocean scenes and debris types.

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